Q15: Context, Retrieval, and Memory

Windows, RAG, and the volatility of context

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The question

Student question (#13, #15)
How do context windows, retrieval, and conversation memory work, and how do we avoid hallucination without deleting context?

The context window + its cost

- All the model "knows now" must fit in n tokens.
- Attention compute is $O(n^2)$; **KV cache** grows linearly in n.
- "Just make it bigger" is *not* free.

Retrieval-augmented generation

embed docs $\rightarrow$ index $\rightarrow$ embed query $\rightarrow$ top-k $\rightarrow$ put in context.

- Draw on a corpus far larger than any window.
- Pay attention cost only on retrieved chunks.
- **Cite grounded sources** $\Rightarrow$ less hallucination.

Long context vs. retrieval

- **Long context**: coherent whole, reason jointly (contract, code file).
- **Retrieval**: vast, sparse, fresh, or must-attribute knowledge.
- Common pattern: retrieve to narrow, then long-window reason.

Why context is not memory

- Context = **volatile** working memory; gone when chat ends / scrolls out.
- Never written back to weights (Q14's long-term store).
- Agent memory: persist to a store, summarize, *re-inject* selectively.

"Don't delete context" = what to persist/summarize/retrieve, not infinite windows.